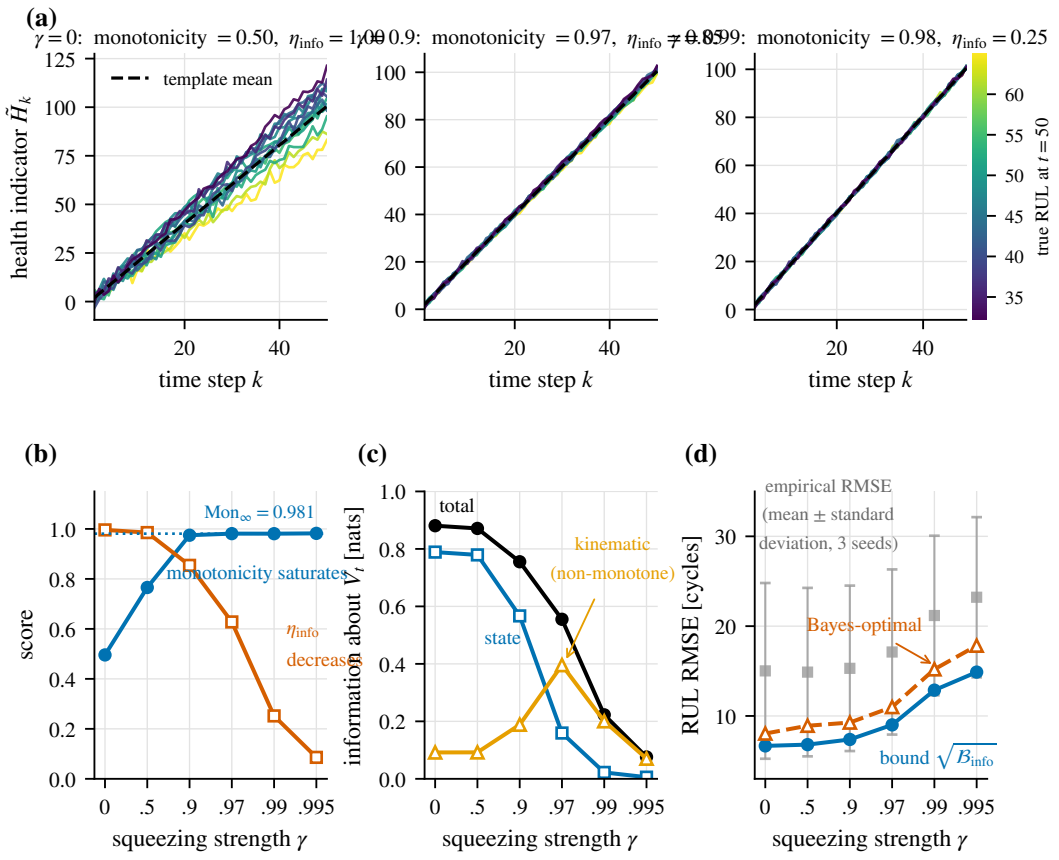
(a)
γ = 0: monotonicity = 0.50, η_info = 1.00
γ = 0.9: monotonicity = 0.97, η_info = 0.85
γ = 0.999: monotonicity = 0.98, η_info = 0.25
template mean
health indicator $\tilde{H}_k$
time step k
true RUL at $t = 50$
(b)
score
Mon$_\infty$ = 0.981
monotonicity saturates
η_{info} decreases
squeezing strength γ
(c)
information about V_t [nats]
total
kinematic (non-monotone)
state
squeezing strength γ
(d)
RUL RMSE [cycles]
empirical RMSE (mean ± standard deviation, 3 seeds)
Bayes-optimal
bound $\sqrt{B_{\text{info}}}$
squeezing strength γ